

HARIKRUSHN GAJJAR

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ABOUT ME

Detail-oriented Computer Science graduate with a strong foundation in full-stack development, real-time systems, and software architecture. Experience building and deploying functional applications from cloud-connected web platforms to hardware-level embedded systems. Adept at optimizing system performance, streamlining deployment workflows, and collaborating effectively within agile development cycles.

EDUCATION

Bachelor's degree in computer science – University of Manitoba, Graduated June 2026

- Key Coursework: Software Engineering, Embedded Systems, Web Development, Operating Systems, Advanced Databases, Parallel Systems.

WORK EXPERIENCE

Inventory tracking webapp: B.A.P.S. organization – TypeScript | React | Firebase | SQL

Apr 2026 – present

- Architected and maintained Relational (SQL) and Firebase databases to track inventory and manage records for big events needing huge inventory.
- Participated in full-cycle software development, writing clean OOP logic, debugging frontend interfaces, and performing unit testing to ensure software reliability.
- Documented system architecture and workflow logic to assist with production support, onboarding, and ongoing software maintenance.
- Explored alternative implementation and optimization workflows for improvement.

Akshar Electric webapp: Akshar Digital Architecture company – TypeScript | React | Firebase

Aug 2025 – Aug 2025 (<https://aksharelectric.ca/>)

- Designed and developed a modern, interactive website using AI assisted tools.
- Implemented major customizations manually to align the web app with client requirements and expectations.
- Identified and resolved errors generated by AI to improve the website's reliability, security, and usability.
- Structured the application with scalability in mind, enabling easier maintenance, future updates, and feature expansion.

ACADEMIC PROJECTS

Embedded Real-Time game System – C | Microcontroller programming

Sep 2025 – Nov 2025

- Implemented hardware-based game using Microchip Curiosity Nano click board with 2x2 button, Joystick, and OLED display.
- Configured microcontroller registers directly for interrupts, timers, and GPIO to ensure strict game logic flow without fail.
- Built a Rate Monotonic Scheduling (RMS) system using a 1 ms timer to increase reliability.
- Optimized OLED rendering from full-screen updates to object-level drawing to improve frame rate from 10 to 30.
- Integrated SPI-based display and input peripherals and implemented an interrupt-driven reset mechanism.
- Used system debugging and created strict requirements to increase reliable usage of the equipment used to run the game.

Server and client-side communication system – Python | JavaScript

May 2025 – June 2025

- Developed a client-server communication system enabling real-time data exchange over network sockets and HTTP.
- Implemented asynchronous I/O handling and request/response protocols, concurrent processing of multiple clients.
- Designed and built a RESTful API and WebSocket interface for communication between the frontend and backend.
- Debugged and logged server behavior to troubleshoot client-side issues.
- Wrote documentation to ensure proper understanding of the system and its components.

LANGUAGES: Python, Java, C/C++, Javascript, Typescript, Golang

Frameworks & Web: HTML5/CSS, React, RESTful APIs, WebSockets

Cloud & Databases: Firebase, HSQLDB, SQL, JSON

Tools: Git, VS Code, IntelliJ, Android Studio, Linux

Methodologies: OOP, SDLC, Unit testing (JUnit), Agile workflows